

Complex Inner Product Spaces

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Introduction

It was the custom of Professor Mac Nerney, both at the University of North Carolina and the University of Houston, to offer his students, upon completion of his course on analytic functions, whose content may be found in [2], a further course. The subject matter of this latter is somewhat hard to title, varying from time to time with the interests and inclinations both of teacher and student. The title of this paper, Complex Inner Product Spaces, must surely come as close as any to describing this subject matter. Here you will find the two themes, each dear to the heart of Mac Nerney, Hellinger Integrals and Kernels in Functional Hilbert Spaces. Here, also, following the pattern of [2], the reader is invited to offer argument for the assertions that are true and counter-example to those that are not.

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Chapter 1

Let K be a function from $[0, 1] \times [0, 1]$ such that $K(u, v) = \frac{u+v-|u-v|}{2} - uv$ for each $\{u, v\}$ in $[0, 1] \times [0, 1]$.

Theorem 1.1 If each of x and y is a function from $[0, 1]$ to the number-plane and x is continuous, then the following statements are equivalent.

- (1) $x = -y^{(2)}$ and $y(0) = y(1) = 0$.
- (2) $y(u) = \int_0^1 K[u, I] x \, dI$ for each u in $[0, 1]$.

Theorem 1.2 If each of $\{x_1, y_1\}$ and $\{x_2, y_2\}$ is an ordered pair of functions from $[0, 1]$ to the plane, each of x_1 and x_2 is continuous, and for each u in $[0, 1]$

$$y_1(u) = \int_0^1 K[u, I] x_1 \, dI \text{ and } y_2(u) = \int_0^1 K[u, I] x_2 \, dI,$$

then

$$\int_0^1 y_1 x_2^* \, dI = \int_0^1 x_1 y_2^* \, dI.$$

Theorem 1.3 If z is a complex number and x is a continuous function from $[0, 1]$ to the plane which is not the constant 0, then the following statements are equivalent.

- (1) $\int_0^1 K[u, I] x \, dI = zx(u)$ for each u in $[0, 1]$.
- (2) There exists a positive integer n and a point c different from 0 such that x is the contraction to $[0, 1]$ of $c S[n\pi I]$ and $z = 1/(n\pi)^2$.

Let z, k , and A be sequences such that for $n = 1, 2, \dots$, $z_n = 1/(\pi n)^2$, k_n is the contraction to $[0, 1]$ of $2^{1/2} S[n\pi I]$, and A_n is a function from $[0, 1] \times [0, 1]$ such that $A_n(u, v) = \sum_1^n z_p k_p(u) k_p(v)^*$; let K_1 be K minus the limit of the sequence A .

Theorem 1.4 The sequence A converges absolutely and uniformly on $[0, 1] \times [0, 1]$ and, if p and q are positive integers, $\int_0^1 |k_p|^2 \, dI = 1$ and $\int_0^1 k_p k_q^* \, dI = 0$.

Theorem 1.5 If w is a complex number and x is a continuous function from $[0, 1]$ to the plane such that $\int_0^1 K_1[u, I] x \, dI = w x(u)$ for each u in $[0, 1]$, then

- (1) if $w \neq 0$, then $\int_0^1 x k_p^* \, dI = 0$ for $p = 1, 2, \dots$, and
- (2) if x is not the constant 0, then $w = 0$.

Theorem 1.6 Suppose n is a positive integer, $\{t_p\}_0^{n+1}$ is an increasing number-sequence with $t_0 = 0$ and $t_{n+1} = 1$, $\{c_p\}_0^n$ is a point sequence, and $y = \sum_1^n K[I, t_p] c_p$. If p is a positive integer less than $n + 1$, then

$$c_p = \frac{y(t_p) - y(t_{p-1})}{t_p - t_{p-1}} - \frac{y(t_{p+1}) - y(t_p)}{t_{p+1} - t_p}.$$

Theorem 1.7 If $b > 0$ and x is a continuous function from $[0, 1]$ to the plane such that $x(0) = x(1) = 0$, there exists a positive integer n and point-sequence $\{c_p\}_0^n$ such that

$$\left| x(u) - \sum_1^n K(u, \frac{p}{n+1}) c_p \right| < b \text{ for each } u \text{ in } [0, 1].$$

Remark With the suppositions of Theorem 1.6,

$$\sum_{p,q=0}^n K(t_p, t_q) c_p^* c_q = \sum_{p=0}^n \frac{|y(t_{p+1}) - y(t_p)|^2}{t_{p+1} - t_p}$$

Chapter 2

Let K be a continuous function from $[0, 1] \times [0, 1]$ to the number-plane such that $K(u, v)^* = K(v, u)$ for each $\{u, v\}$ in $[0, 1] \times [0, 1]$, μ be an increasing function from $[0, 1]$ to a set of numbers, and H be the function to which $\{x, y\}$ belongs only in case each of x and y is a function from $[0, 1]$ to the number-plane and x is continuous and $y(u) = \int_0^1 K[u, I] x d\mu$ for each u in $[0, 1]$; the statement that $y = Hx$ means that $\{x, y\}$ belongs to H .

Theorem 2.1 If each of x_1 and x_2 is a continuous function from $[0, 1]$ to the number-plane, then each of Hx_1 and Hx_2 is continuous and $\int_0^1 (Hx_1) x_2^* d\mu = \int_0^1 x_1 (Hx_2)^* d\mu$.

Let $\mathbf{S}$ be the set of all continuous functions from $[0, 1]$ to the number-plane and Q be a function from $\mathbf{S} \times \mathbf{S}$ such that $Q(x, y) = \int_0^1 x y^* d\mu$ for each $\{x, y\}$ in $\mathbf{S} \times \mathbf{S}$.

Theorem 2.2 If each of x and y is in $\mathbf{S}$, then

$$Q(Hx, y) = Q\left(H \frac{x+y}{2}, \frac{x+y}{2}\right) - Q\left(H \frac{x-y}{2}, \frac{x-y}{2}\right) + iQ\left(H \frac{x+iy}{2}, \frac{x+iy}{2}\right) - iQ\left(H \frac{x-iy}{2}, \frac{x-iy}{2}\right).$$

Theorem 2.3 If $Q(Hx, x) = 0$ for each x in $\mathbf{S}$ then, for each x in $\mathbf{S}$, Hx is the constant 0 on $[0, 1]$; hence, in this case, K is the constant 0 on $[0, 1] \times [0, 1]$.

Theorem 2.4 If each of x and y is in $\mathbf{S}$, then $Q(x - ry, x - ry) \geq 0$ for each number r , and therefore $|Q(x, y)|^2 \leq Q(x, x) Q(y, y)$.

Theorem 2.5

(1) There are numbers N_1 and N_2 such that if $\{x, y\}$ belongs to H then $|y(u)|^2 \leq N_1 Q(x, x)$ for each u in $[0, 1]$ and $|Q(Hx, x)| \leq N_2 Q(x, x)$.

(2) For each positive number c there is a positive number b such that, if $\{x, y\}$ is in H and $Q(x, x) \leq 1$ and $[u, v]$ lies in $[0, 1]$ and $v - u \leq b$, then $|y(v) - y(u)| \leq c$.

Let m and M denote, respectively, the greatest lower bound and the least upper bound of the set to which w belongs only in case there is an x in $\mathbf{S}$ such that $Q(x, x) = 1$, $Q(Hx, x) = w$.

Theorem 2.6 If z is a complex number and there is a member x of $\mathbf{S}$ which is not the constant 0 on $[0, 1]$ and $Hx = zx$, then z is a number and $m \leq z \leq M$.

Theorem 2.7 If $\{x, y\}$ is in $\mathbf{S} \times \mathbf{S}$, $m Q(x - ry, x - ry) \leq Q(H(x - ry), x - ry) \leq M Q(x - ry, x - ry)$. Therefore

$$|MQ(x, y) - Q(Hx, y)|^2 \leq \{MQ(x, x) - Q(Hx, x)\} \{MQ(y, y) - Q(Hy, y)\}$$

and

$$|Q(Hx, y) - mQ(x, y)|^2 \leq \{Q(Hx, x) - mQ(x, x)\} \{Q(Hy, y) - mQ(y, y)\}.$$

Theorem 2.8 Suppose x is an infinite sequence with final set lying in $\mathbf{S}$ such that the number-sequence $Q(Hx, x)$ has limit M and, for each non-negative integer p , $Q(x_p, x_p) = 1$.

(1) There is an infinite subsequence $\bar{x}$ of x and a member X of $\mathbf{S}$ such that the function-sequence $H\bar{x}$ converges uniformly and has limit X .

(2) For each nonnegative integer p , $|Q(Hx_p, x_p)|^2 \leq Q(Hx_p, Hx_p)$ so that $Q(X, X) \geq M^2$.

(3) If y is in $\mathbf{S}$ then, for each nonnegative integer p ,

$$|MQ(Hx_p, y) - Q(HHx_p, y)|^2 \leq \{M - Q(Hx_p, x_p)\} \{MQ(Hy, Hy) - Q(HHy, Hy)\}.$$

(4) $MQ(X, y) = Q(HX, y)$ for each y in $\mathbf{S}$, so that $HX = MX$.

Theorem 2.9 There is a member Y of $\mathbf{S}$ such that $Q(Y, Y) \geq m^2$ and $HY = mY$.

The following corollaries apply to Chapter 1.

Corollary 1 The function K_1 has only the value 0. What is the value of $\sum_1^\infty z_p$?

Corollary 2 If x is a continuous function from $[0, 1]$ to the number-plane such that the function-sequence $\sum_1^\infty (\int_0^1 x k_p^* dI) k_p$ converges uniformly and has limit y on $[0, 1]$, then $\int_0^1 K[u, I] \{x - y\} dI = 0$ for each u in $[0, 1]$ and therefore $y = x$.

Corollary 3 If x is a continuous function from $[0, 1]$ to the number-plane such that $x(0) = x(1) = 0$ and $c > 0$, there exists a positive integer n and a point-sequence $\{B_p\}_0^n$ such that $|x(u) - \sum_0^n B_p S(p\pi u)| < c$ for each u in $[0, 1]$.

Corollary 4 If y is a continuous function from the number-interval $[a, b]$ to the number-plane and $c > 0$, there exists a positive integer n and a point-sequence $\{C_p\}_0^n$ such that $|y(u) - \sum_0^n C_p u^p| < c$ for each u in $[a, b]$.

Remark If K_2 is the function from $[0, 1] \times [0, 1]$ defined by $K_2(u, v) = \mu \left(\frac{u + v - |u - v|}{2} \right)$, $\{t_p\}_0^n$ is an increasing number-sequence with $t_0 = 0$ and $t_n = 1$, $\{c_p\}_0^n$ is a point-sequence and $y = \sum_0^n K_2 [I, t_p] c_p$, then

$$\sum_{p, q=0}^n K_2(t_p, t_q) c_p^* c_q = |y(0)|^2 \mu(0) + \sum_{p=0}^n \frac{|y(t_p) - y(t_{p-1})|^2}{\mu(t_p) - \mu(t_{p-1})}.$$

Exempli Gratia

Let S' denote the set to which x belongs only in case x is an analytic function with initial set the unit disk U and there is a number b such that if r is a number in $(0, 1)$, then $\int_{-\pi}^{\pi} |x[rE[iI]]|^2 dI < b$. There is a function Q from $S' \times S'$ to the number-plane such that, if $\{x, y\}$ is in $S' \times S'$ and $c > 0$, there is a number u in $(0, 1)$ such that if r is a number in $(u, 1)$, then $|Q(x, y) - \frac{1}{2\pi} \int_{-\pi}^{\pi} x[rE[iI]] y[rE[iI]]^* dI| < c$; Q has the property shown in Theorem

2.4. The function k , from $U \times U$, defined by $k(s, t) = \frac{1}{1 - t^*s}$, has the property that if t is in U , then $k[I, t]$ is in S' and $Q(x, k[I, t]) = x(t)$ for each x in S' .

The subset H of $S' \times S'$, to which $\{x, y\}$ belongs in case $y(0) = x(0)$ and for each t in U different from 0, $y(t) = \frac{1}{t} \int_{[0;1]} x$, is a function from S' into S' such that

(1) $Q(x, Hy) = Q(Hx, y)$ for each $\{x, y\}$ in $S' \times S'$, and

(2) if x is an infinite sequence with final set lying in S' and, for each nonnegative integer p , $Q(x_p, x_p) \leq 1$, then there is a subsequence $\bar{x}$ of x and a member X of S' such that the number-sequence $Q(X - H\bar{x}, X - H\bar{x})$ has limit 0.

Chapter 3

Suppose Σ is a linear family of functions, in the sense that if each of x and y is in Σ and c is a complex number, then each of $x+y$ and cx is in Σ (with the tacit assumption that each two functions in Σ have the same initial set). Suppose Q is a function from $\Sigma \times \Sigma$ to the plane such that

- (i) $Q(x+y, z) = Q(x, z) + Q(y, z)$ for all x, y , and z in Σ ,
- (ii) $Q(cx, y) = cQ(x, y)$ for all x and y in Σ and each complex number c ,
- (iii) $Q(x, y)^* = Q(y, x)$ for all x and y in Σ , and
- (iv) $Q(x, x) > 0$ for all x in Σ , other than 0.

Define $N(x)$ to be $Q(x, x)^{1/2}$ for each x in Σ , and suppose H is a function from Σ into Σ such that

- (1) $Q(x, Hy) = Q(Hx, y)$ for each $\{x, y\}$ in $\Sigma \times \Sigma$, and
- (2) if u is an infinite sequence with final set lying in Σ and the number-sequence $N(u)$ has bounded final set, then there is a subsequence v of u and a member X of Σ such that the number-sequence $N(X - Hv)$ has limit 0.

Theorem 3.1 If each of x and y is in Σ then $|Q(x, y)|^2 \leq Q(x, x)Q(y, y)$ and $N(x+y) \leq N(x) + N(y)$; moreover, H is linear — in the sense that if each of $\{x_1, y_1\}$ and $\{x_2, y_2\}$ is in H and c is a complex number, then each of $\{x_1 + x_2, y_1 + y_2\}$ and $\{cx_1, cy_1\}$ is in H .

Theorem 3.2 There exists a number b such that $N(Hx) \leq bN(x)$ for each x in Σ .

Theorem 3.3 If each of x and y is in Σ then

$$Q(Hx, y) = Q\left(H\frac{x+y}{2}, \frac{x+y}{2}\right) - Q\left(H\frac{x-y}{2}, \frac{x-y}{2}\right) + iQ\left(H\frac{x+iy}{2}, \frac{x+iy}{2}\right) - iQ\left(H\frac{x-iy}{2}, \frac{x-iy}{2}\right)$$

Theorem 3.4 If $b > 0$, then each two of the following statements are equivalent:

- (1) $N(Hx) \leq bN(x)$ for each x in Σ .
- (2) $|Q(Hx, x)| \leq bQ(x, x)$ for each x in Σ .
- (3) $|Re Q(Hx, y)| \leq bN(x)N(y)$ for each $\{x, y\}$ in $\Sigma \times \Sigma$.
- (4) $|Q(Hx, y)| \leq bN(x)N(y)$ for each $\{x, y\}$ in $\Sigma \times \Sigma$.

Let B denote the least nonnegative number b such that if x is in Σ then $N(Hx) \leq bN(x)$.

Theorem 3.5 If z is a complex number and there is a non-trivial member u of Σ such that $Hu = zu$, then z is a number and $|z| \leq B$.

Theorem 3.6 If z_1 and z_2 are numbers, each of u_1 and u_2 is a non-trivial member of Σ , and $Hu_1 = z_1u_1$ and $Hu_2 = z_2u_2$, then $Q(u_1, u_2) = 0$.

Theorem 3.7 Suppose $H \neq 0$ so that $B > 0$. There is a number z and a member u of Σ such that $|z| = B$ and $N(u) = 1$ and $Hu = zu$.

Theorem 3.8 Suppose $\{u_j\}_0^n$ is a sequence with final set lying in Σ and if each of j and k is a nonnegative integer not greater than n , then $N(u_j) = 1$ and $Q(u_j, u_k) = 0$ unless $j = k$,

and P is the function from Σ defined by : $Px = \sum_0^n Q(x, u_j)u_j$ for each x in Σ .

(1) P has both the properties postulated for H , and $P^2 = P$, i. e., if x is in Σ , then $PPx = Px$.

(2) In order that $Q(y, u_k) = 0$ for each nonnegative integer k not greater than n , it is necessary and sufficient that there be a member x of Σ such that $y = x - Px$.

Theorem 3.9 One of the following alternatives holds:

(I) There is a number z and a member u of Σ such that $N(u) = 1$ and if x is in Σ , then $Hx = zQ(x, u)u$.

(II) There exist a positive integer n , a number-sequence $\{z_j\}_0^n$, and a sequence $\{u_j\}_0^n$ with final set lying in Σ such that

- (a) $|z_0| = 1$. u. b. $N(Hx)$ for x in Σ such that $N(x) \leq 1, N(u_0) = 1$, and $Hu_0 = z_0u_0$.
- (b) if k is a nonnegative integer less than n , $|z_{k+1}| = 1$. u. b. $N(H[x - \sum_0^k Q(x, u_j)u_j])$ for x in Σ such that $N(x - \sum_0^k Q(x, u_j)u_j) \leq 1, Q(u_{k+1}, u_j) = 0$ for each nonnegative integer j not greater than $k, N(u_{k+1}) = 1$, and $Hu_{k+1} = z_{k+1}u_{k+1}$, and
- (c) if x in Σ and n is a nonnegative integer, then $Hx = \sum_0^n z_j Q(x, u_j)u_j$.

(III) There exists an infinite number-sequence z with final set not containing 0 and an infinite sequence u with final set lying in Σ such that

- (a) $|z_0| = 1$. u. b. $N(Hx)$ for x in Σ such that $N(x) \leq 1, N(u_0) = 1$, and $Hu_0 = z_0u_0$.
- (b) if k is a nonnegative integer, $|z_{k+1}| = 1$. u. b. $N(H[x - \sum_0^k Q(x, u_j)u_j])$ for x in Σ such that $N(x - \sum_0^k Q(x, u_j)u_j) \leq 1, Q(u_{k+1}, u_j) = 0$ for each nonnegative integer j not greater than $k, N(u_{k+1}) = 1$, and
- (c) if x is in Σ and n is a nonnegative integer, then

$$N(Hx - \sum_0^n z_j Q(x, u_j)u_j) \leq |z_{n+1}| N(x - \sum_0^n Q(x, u_j)u_j).$$

Theorem 3.10 Supposing 3.9 (III), the following are true:

- (1) If j and k are nonnegative integers, then $N(Hu_j - Hu_k)^2 = z_j^2 + z_k^2$.
- (2) The sequence z has limit 0.

Theorem 3.11 Supposing 3.9 (III), there exist a reversible number-sequence λ , such that $|\lambda|$ is nonincreasing and has limit 0, and an infinite sequence P , each value of which is a function of the type introduced in Theorem 3.8, such that the following hold:

- (1) If j and k are nonnegative integers, $Q(P_jx, P_ky) = 0$ for each $\{x, y\}$ in $\Sigma \times \Sigma$.
- (2) If j is a nonnegative integer, then $HP_jx = \lambda_j P_jx$ for each x in Σ .
- (3) For each nonnegative integer n and each member x of Σ ,

$$N(Hx - \sum_0^n \lambda_j P_jx) \leq |\lambda_{n+1}| N(x - \sum_0^n P_jx) \leq |\lambda_{n+1}| N(x).$$

The following corollaries apply to Chapter 2, with the identification of Σ as the set $\mathbf{S}$, as in that chapter, of all continuous functions from $[0, 1]$ to the plane with, as there, for each

$\{x, y\}$ in $\mathbf{S} \times \mathbf{S}$,

$$Q(x, y) = \int_0^1 x y^* d\mu.$$

Corollary 1 Alternatives (I) and (II) of Theorem 3.9 correspond to the form

$$K(s, t) = \sum_0^n z_j u_j(s) u_j(t)^*.$$

With the supposition of alternative (III) in Theorem 3.9, we have the following:

Corollary 2 Let N_1 be as indicated in Theorem 2.5(1).

(1) If c is an infinite point-sequence and k and n are nonnegative integers,

$$\left| \sum_{k+1}^{k+n} z_j c_j u_j(t) \right|^2 \leq N_1 \sum_{k+1}^{k+n} |c_j|^2 \text{ for all } t \text{ in } [0, 1].$$

(2) If x is in Σ , then $\sum_0^n \left| \int_0^1 x u_j^* d\mu \right|^2 \leq \int_0^1 |x|^2 d\mu$ for each nonnegative n .

(3) If x is in Σ , the function-sequence $\sum_0^\infty z_j Q(x, u_j) u_j$ converges uniformly on $[0, 1]$.

(4) If x is in Σ then, for each nonnegative integer n ,

$$\int_0^1 \left| Hx - \sum_0^n z_j Q(x, u_j) u_j \right|^2 d\mu \leq z_{n+1}^2 \int_0^1 |x|^2 d\mu,$$

so that $\sum_0^\infty z_j Q(x, u_j) u_j$ has limit Hx on $[0, 1]$.

Corollary 3 For each $\{x, y\}$ in $\Sigma \times \Sigma$, $\sum_0^\infty z_j Q(x, u_j) Q(u_j, y)$ has limit $\int_0^1 (Hx) y^* d\mu$.

Corollary 4 Let $K^{(2)}$ be the function from $[0, 1] \times [0, 1]$ defined by

$$K^{(2)}(s, t) = \int_0^1 K[s, I] K[I, t] d\mu.$$

(1) If t is in $[0, 1]$, $\sum_0^\infty z_j u_j(t)^* u_j$ has the limit $K^{(2)}[I, t]$ uniformly on $[0, 1]$.

(2) $\sum_0^\infty z_j^2 |u_j|^2$ has the limit $K^{(2)}[I, I]$ uniformly on $[0, 1]$, so that $\sum_0^\infty z_j^2$ is convergent, with limit $\int_0^1 K^{(2)}[I, I] d\mu$.

(3) If x is in Σ , then $\sum_0^\infty |z_j Q(x, u_j) u_j|$ converges uniformly on $[0, 1]$.

(4) If t is in $[0, 1]$ and $\sum_0^\infty z_j u_j(t)^* u_j$ converges uniformly on $[0, 1]$, then it has limit $K[I, t]$.

Remark There is the further result (Mercer's Theorem) regarding alternative (III), to the effect that if the sequence z is positive, then $\sum_0^\infty z_j u_j(s) u_j(t)^*$ converges uniformly for $\{s, t\}$ in $[0, 1] \times [0, 1]$: hence, the following conditions are equivalent.

(1) $Q(Hx, x) \geq 0$ for each x in Σ .

(2) The sequence λ is nonnegative.

(3) $\sum_{p,q=0}^n K(t_p, t_q) c_p^* c_q \geq 0$ for each sequence $\{t_p\}_0^n$ with final set lying in $[0, 1]$ and each complex number-sequence $\{c_p\}_0^n$.

Definition If Σ is a linear family of functions, a function Q from $\Sigma \times \Sigma$ to the plane with properties (i), (ii), (iii), and (iv) from the beginning paragraph of this chapter is an inner product for Σ , the ordered pair $\{\Sigma, Q\}$ is an inner product space, and the function N from Σ to the numbers such that if x is in Σ then $N(x) = Q(x, x)^{1/2}$ is the norm (for Σ) corresponding to Q .

Chapter 4

Project 4.1 (Euclidian Spaces) Let P be a positive integer, Π the set of all integers in $[0, P]$, S the set of all functions from Π to the number-plane, and Q be a function from $S \times S$ such that if $\{x, y\}$ is in $S \times S$ then $Q(x, y) = \sum_{s \in \Pi} x(s)y(s)^*$. The inner product space $\{S, Q\}$ is complete: If N is the norm corresponding to Q then each sequence (with final set lying in S) which converges with respect to N has a limit with respect to N . Suppose K is a function from $\Pi \times \Pi$ to the number-plane such that $K(s, t)^* = K(t, s)$ for each $\{s, t\}$ in $\Pi \times \Pi$ and let H be the function from S to S to which $\{x, y\}$ belongs only in case $y(t) = Q(x, K[I, t])$ for each t in Π . Theorem 3.9 is applicable here.

Project 4.2 Suppose R is a collection of subsets of the set Ω ; two members of R overlap in case their common part includes a member of R . Let D be the collection of all finite subsets M of R such that no two members of M overlap and M^* is in R : a partitioning of the member A of R is a member M of D such that $M^* = A$, and a refinement of such an M is a partitioning N of A each member of which is a subset of a member of M . Suppose that, for each two members A and B of R ,

(1) there is a partitioning M of B , each member of which either lies in A or does not overlap A , such that if B overlaps A then the common part of A and B is the star of the subset of M to which G belongs only in case G lies in A , and

(2) there is a member of D which includes a partitioning of A and a partitioning of B . Suppose, finally, that ω is a function from R to the nonnegative numbers and that ω is additive, that is, if M is in D then $\omega(M^*) = \sum_{G \in M} \omega(G)$. Let S be the set to which x belongs only in case x is a function from R to the number-plane, x is additive, if A is in R and $\omega(A) = 0$ then $x(A) = 0$, there is a number b such that if M is in D then $\sum_{G \in M} \frac{|x(G)|^2}{\omega(G)} \leq b$ — here,

we adopt the convention that if $\omega(G) = 0$ then $\frac{|x(G)|^2}{\omega(G)}$ denotes the number 0. If x is in S

and $\int_R \frac{|x|^2}{\omega}$ denotes the least number b such that if M is in D then $\sum_{G \in M} \frac{|x(G)|^2}{\omega(G)} \leq b$, then for each positive number c there exists a member M of D such that if N is a member of

D which includes a refinement of M then $\int_R \frac{|x|^2}{\omega} - c < \sum_{H \in N} \frac{|x(H)|^2}{\omega(H)}$. There is an inner

product Q for S such that if $\{x, y\}$ is in $S \times S$ then $Q(x, y) = \int_R \frac{xy^*}{\omega}$, that is, if $\{x, y\}$ is in $S \times S$ and $c > 0$, there is a member M of D such that if N is a member of D which includes a refinement of M then $\left| Q(x, y) - \sum_{H \in N} \frac{x(H)y(H)^*}{\omega(H)} \right| < c$ — with the obvious convention (as before) if $\omega(H) = 0$.

There is a function K from $R \times R$ to the numbers such that, for each $\{s, t\}$ in $R \times R$, if s does not overlap t then $K(s, t) = 0$ but if s overlaps t and M is a partitioning of s , each member of which lies in t or does not overlap t , such that the common part of s and t is the star of the subset N of M to which H belongs only in case H lies in t , then $K(s, t) = \sum_{H \in N} \omega(H)$. If t is in R then the function $K[\cdot, t]$, to which $\{s, p\}$ belongs only in case s is in R and $p = K(s, t)$, belongs to S and $Q(x, K[\cdot, t]) = x(t)$ for each member x of S . The inner product space $\{S, Q\}$ is complete.

Project 4.3 As an instance of the space $\{S, Q\}$ of Project 4.2, let R be the collection of

all finite subsets of Ω and, for each G in R , $\omega(G)$ be 1 in case G is degenerate but $\omega(G)$ be the number of members of G otherwise. For each function x from Ω to the number-plane, let dx be a function from R such that if H is in R then $dx(H) = \sum_{t \in H} x(t)$. Let S_1 be the set to which x belongs only in case x is a function from Ω to the plane such there is a number b such that, if F is a finite subset of Ω , $\sum_{t \in F} |x(t)|^2 \leq b$. The inner product Q_1 , such that if $\{x, y\}$ is in $S_1 \times S_1$ then $Q_1(x, y) = Q(dx, dy)$, has the following characterization: if $\{x, y\}$ is in $S_1 \times S_1$ and $c > 0$, there is a finite subset F of Ω such that $|Q_1(x, y) - \sum_{s \in G} x(s)y(s)^*| < c$ for each finite subset G of Ω which includes F . The space $\{S_1, Q_1\}$ is sometimes denoted by $\ell^2(\Omega)$; if Ω is the set of all nonnegative integers, the customary notation is simply ℓ^2 , called (complex) Hilbert Space.

The notation $\sum_{s \in \Omega} x(s)y(s)^*$ is understood to mean $Q_1(x, y)$ for each $\{x, y\}$ in $S_1 \times S_1$. The Kronecker's function δ (for Ω) is defined on $\Omega \times \Omega$ in such a way that $\delta(s, t)$ is 1 or 0 according as s is t or s is not t .

Project 4.4 (Hellinger integrals) As another instance of the space $\{S, Q\}$ of Project 4.2, let Ω be the number interval $[A, B]$ and R be the collection of all subintervals of Ω . For each function x from Ω to the plane let dx be the function from R such that if $[a, b]$ is in R then $dx([a, b]) = x(b) - x(a)$. Suppose μ is a nondecreasing non-constant function from Ω to the numbers, and let $\omega = d\mu$.

If each of x and y is a function from $[A, B]$ to the plane such that $\{dx, dy\}$ is in $S \times S$, then each of x and y is Hellinger integrable from A to B and $Q(dx, dy)$ is customarily denoted by $\int_A^B \frac{dx dy^*}{d\mu}$.

If S_1 is a linear family of functions x from $[A, B]$ to the plane such that (1) dx is in S and (2) dx is the constant function 0 on R only in case x is the constant 0 on $[A, B]$, then there is an inner product Q_1 for S_1 such that $Q_1(x, y) = Q(dx, dy)$ for $\{x, y\}$ in $S_1 \times S_1$.

An example of S_1 such that $\{S_1, Q_1\}$ is complete arises in the context of Chapter 1. However, we will not pursue it here. Rather, we will present the following example. Suppose that λ is a continuous function from $[A, B]$ to the numbers such that $\lambda(A) \neq \lambda(B)$, and let S_1 be the set of all x from $[A, B]$ to the plane such that dx is in S and $\int_A^B x d\lambda = 0$, Q_1 defined as above.

Let $C = \int_A^B \frac{d\lambda[A, I]}{d\lambda(A, B)} \frac{d\lambda[I, B]}{d\lambda(A, B)} d\mu$ and K_1 a function from $[A, B] \times [A, B]$ such that $K_1(s, t)$ is

$$\int_A^s \frac{d\lambda[A, I]}{d\lambda(A, B)} d\mu + \int_t^B \frac{d\lambda[I, B]}{d\lambda(A, B)} d\mu - C$$

or

$$\int_A^t \frac{d\lambda[A, I]}{d\lambda(A, B)} d\mu + \int_s^B \frac{d\lambda[I, B]}{d\lambda(A, B)} d\mu - C$$

according as $s \leq t$ or $t \leq s$, for each $\{s, t\}$ in $[A, B] \times [A, B]$. If T is in $[A, B]$ then $K_1[I, t]$ is in S_1 and $Q_1(x, K_1[I, t]) = x(t)$ for each member x of S_1 ; $\{S_1, Q_1\}$ is complete.

Project 4.5 (the Hardy class H^2) Let S be the set to which x belongs only in case x an analytic function from U to the plane and there is a number b such that if r is in $(0, 1)$ then $\int_{-\pi}^{\pi} |x[rE[iI]]|^2 dI < b$, and for each $\{x, y\}$ in $S \times S$ let $h_{x,y}$ be the function from $(0, 1)$ defined by $h_{x,y}(r) = \int_{-\pi}^{\pi} x[rE[iI]] y[rE[iI]]^* dI$. We have the inner product Q_1 for S defined by $Q_1(x, y) = \frac{1}{2\pi} (h_{x,y}(1-)) = \lim_{\substack{r \rightarrow 1 \\ r < 1}} h_{x,y}(r)$, the function K from $U \times U$ defined

by $K(s, t) = \frac{1}{1 - st^*}$, and the function H from S to S to which $\{x, y\}$ belongs only in case $y(0) = x(0)$ and $(Iy)' = x$ [see the terminal example in Chapter 2].

There is an inner product Q_2 for S defined by $Q_2(x, y) = \frac{1}{\pi} \int_{0+}^{1-} I h_{x,y} dI$, that is, if $\{x, y\}$ is in $S \times S$ and $c > 0$, there is an interval $[v, V]$ lying in $(0, 1)$ such that if $[r, R]$ lies in $(0, 1)$ and includes $[v, V]$ then $\left| Q_2(x, y) - \frac{1}{\pi} \int_r^R I h_{x,y} dI \right| < c$; $Q_2(x, y) = Q_1(Hx, y)$ for each $\{x, y\}$ in $S \times S$.

Elementary calculus considerations suggest the notation $Q_2(x, y) = \frac{1}{\pi} \iint_U x y^*$, whence we have the formula $Hx(t) = \frac{1}{\pi} \iint_U K[t, I] x$ for each $\{t, x\}$ in $U \times S$; one might note that for $0 < r < R < 1$ and $L(r) \leq s \leq L(R)$, the integral-transformation formulas

$$\begin{aligned} \int_r^R I h_{x,y} dI &= \int_{L(r)}^{L(R)} E h_{x,y}[E] dE = \int_{L(r)}^{L(R)} E^2 h_{x,y}[E] dI, \\ E(s)^2 h_{x,y}(E(s)) &= \int_{-\pi}^{\pi} |E[s + iI]|^2 x[E[s + iI]] y[E[s + iI]]^* dI \\ &= \int_{-\pi}^{\pi} \{(x[E]E')(y[E]E')^*\} [s + iI] dI, \text{ and} \\ \iint_{E([L(r), L(R)] \times [-\pi, \pi])} xy^* &= \iint_{E([L(r), L(R)] \times [-\pi, \pi])} (x[E]E')(y[E]E')^*. \end{aligned}$$

The inner product space $\{S, Q_1\}$ is complete, but $\{S, Q_2\}$ is not complete.

Project 4.6 (Fourier series) In the space $\{S_1, Q_1\}$ of Project 4.4, let S_2 be the subset of S_1 to which x belongs only in case $x(A) = x(B)$ and let Q_2 be the contraction of Q_1 to $S_2 \times S_2$. The member x of S_1 belongs to S_2 only in case $Q_1(x, K_1[I, B] - K_1[I, A]) = 0$; $Q_1(K_1[I, B] - K_1[I, A], K_1[I, B] - K_1[I, A]) = d\mu(A, B)$.

Let K_2 be the function from $[A, B] \times [A, B]$ such that

$$K_2(s, t) = K_1(s, t) - \frac{1}{d\mu(A, B)} \{K_1(s, B) - K_1(s, A)\} \{K_1(B, t) - K_1(A, t)\}$$

for each $\{s, t\}$ in $[A, B] \times [A, B]$; if t is in $[A, B]$ then $K_2[I, t]$ is in S_2 and $Q_2(x, K_2[I, t]) = x(t)$ for each member x of S_2 . Assume, now, the special case that $\lambda = \mu = I$ and $B = -A =$

π . If $\{s, t\}$ is in $[-\pi, \pi] \times [-\pi, \pi]$, then $K_2(s, t) = \frac{\pi}{6} + \frac{1}{4\pi}(s-t)^2 - \frac{1}{2}|s-t|$.

Let H be the function defined as follows: If x is a continuous function from $[-\pi, \pi]$ to the plane and $\int_{-\pi}^{\pi} x \, dI = 0$, then Hx is a function from $[-\pi, \pi]$ such that $Hx(t) = \int_{-\pi}^{\pi} K_2[t, I] x \, dI$ for each t in $[-\pi, \pi]$. If x is in the initial set of H then, for each number t in $[-\pi, \pi]$, $H(x)' = -\frac{1}{2\pi} \int_{-\pi}^{\pi} I x \, dI - \int_{-\pi}^t x \, dI$, Hx belongs to S_2 , and if y belongs to S_2 then $Q_2(Hx, y) = \int_{-\pi}^{\pi} x y^* \, dI$.

If x is in the initial set of H then Hx is the only function y from $[-\pi, \pi]$ to the plane such that $x = -y^{(2)}$, $y(-\pi) = y(\pi)$, $y'(-\pi) = y'(\pi)$, and $\int_{-\pi}^{\pi} y \, dI = 0$. If z is a complex number and u is a nontrivial member of S_2 such that $Hu = zu$ then there exists a positive integer n and an ordered point-pair $\{h, k\}$ such that $z = (1/n)^2$ and $u = hE[-inI] + kE[inI]$. With S identified as the initial set of H and Q defined by $Q(x, y) = \int_{-\pi}^{\pi} x y^* \, dI$ for each $\{x, y\}$ in $S \times S$, Theorem 3.11 applies where for each positive integer n , $\lambda_{n-1} = (1/n)^2$ and P_{n-1} is the function from S to S such that

$$P_{n-1} x(t) = \frac{1}{2\pi} Q(x, E[-inI])E(-int) + \frac{1}{2\pi} Q(x, E[inI])E(int)$$

for each x in S and t in $[-\pi, \pi]$. Adopting the convention that $\$$ and $\mathbb{C}$ denote the sine and cosine respectively, the following is also a formula for P_{n-1} .

$$P_{n-1} x(t) = \frac{1}{\pi} Q(x, \$[nI])\$(nt) + \frac{1}{\pi} Q(x, \mathbb{C}[nI])\mathbb{C}(nt).$$

If x is a function from $[-\pi, \pi]$ to the plane such that $\int_{-\pi}^{\pi} x \, dI$ exists, the Fourier series of x is the infinite sequence F such that, for each nonnegative integer n , F_n is a function from $[-\pi, \pi]$ to the plane and F_0 is the constant $\frac{1}{2\pi} \int_{-\pi}^{\pi} x \, dI$ and if $n > 0$ and t is in $[-\pi, \pi]$ then

$$\begin{aligned} F_n(t) &= \sum_{p=1}^n \frac{1}{2\pi} E ipt \int_{-\pi}^{\pi} x E[-ipI] \, dI \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} x \, dI + \sum_{p=1}^n \frac{1}{\pi} \{ \$ (pt) \int_{-\pi}^{\pi} x \$[pI] \, dI + \mathbb{C}(pt) \int_{-\pi}^{\pi} x \mathbb{C}[pI] \, dI \}. \end{aligned}$$

Project 4.7 (the Riemann mappings) Let Ω be a region in the plane and S_{Ω} be the set to which x belongs only in case x is an analytic function with initial set Ω and $\iint_{\Omega} |x|^2$ exists. There is an inner product Q_{Ω} for S_{Ω} such that $Q_{\Omega} = \iint_{\Omega} x y^*$ for each $\{x, y\}$ in $S_{\Omega} \times S_{\Omega}$.

Suppose f is a reversible analytic function from Ω onto D , and let $\{S_D, Q_D\}$ be the corresponding inner product for D . The function V , to which $\{x, X\}$ belongs only in case x is in S_{Ω} and $X = (f^{-1})' \cdot x[f^{-1}]$, has the property that if $\{x, X\}$ belongs to it then $x = f' \cdot X[f]$; V is an isomorphism from $\{S_{\Omega}, Q_{\Omega}\}$ onto $\{S_D, Q_D\}$, that is, V is a reversible linear function

from S_Ω onto S_D such that $Q_D(Vx, Vy) = Q_\Omega(x, y)$ for each $\{x, y\}$ in $S_\Omega \times S_\Omega$.

Suppose K_D is a function from $D \times D$ such that if t is in D then $K_D[I, t]$ is in S_D and $Q_D(X, K_D[I, t]) = X(t)$ for each X in S_D , and $\{A, B\}$ is a member of $\Omega \times D$ at which f has slope 1: $V^{-1}K_D[I, B]$ is that member h of S_Ω such that $Q_\Omega(x, h) = x(A)$ for each x in S_Ω . Suppose $B = 0$ and D is the disk $D_b(0)$, so that the region Ω is simple: $K_D(s, t) = \frac{1}{\pi} \left\{ \frac{b}{b^2 - s t^*} \right\}^2$ for each $\{s, t\}$ in $D \times D$; $V^{-1}K_D[I, 0] = f'/\pi b^2$; if x is in S_Ω and $x(A) = 1$, then $Q_\Omega(x, x) = Q_\Omega(x - f', x - f') + \pi b^2$; if g is a reversible analytic function with initial set Ω and slope 1 at $\{A, 0\}$ and the area of $g(\Omega)$ exists, then the area of $g(\Omega)$ exceeds the area πb^2 of $f(\Omega)$ unless g is f .

Chapter 5

Elementary Arithmetic of Kernel Systems

The following three results are abstracted from Part III, subtitled The Linear Span of a Family of Functional Hilbert Spaces, of [1], in particular Theorems 2, 3, and the corollary to the latter theorem.

A kernel-system is a sequence $\{K, R, S, Q\}$, such that $\{S, Q\}$ is a complete inner product space of functions from R to the plane and K is a function from $R \times R$ to the plane such that for t in R , $K(\cdot, t)$ is in S and, for each f in S , $Q(f, K(\cdot, t)) = f(t)$.

Terminology. In a kernel-system, $\{K, R, S, Q\}$, the function K is the evaluation kernel (or reproducing kernel, or kernel function, or kernel) in the space $\{S, Q\}$ and the functions of the form $\sum_{t \in M} K(\cdot, t)x(t)$ (for functions x from finite subsets M of R to the plane) are called K-polygons. The latter terminology seems appropriate from the observation that, for $K(s, t) = 1 + \inf\{s, t\}$ on $[0, 1] \times [0, 1]$, each function f on $[0, 1]$, polygonal in the usual sense, may be represented in this form : one has

$$f = \sum_0^n K(\cdot, u_p)x_p \text{ for some increasing number-sequence,}$$

$$\{u_p\}_0^n \text{ with } u_0 = 0 \text{ and } u_n = 1, f(u_0) = \sum_0^n x_p,$$

$$\text{and } f(u_j) - f(u_{j-1}) = (u_j - u_{j-1}) \sum_j^n x_p \quad (j = 1, \dots, n).$$

Theorem 5.1 If R is a set and K is a function from $R \times R$ to the plane then, in order that there should exist a complete inner product space $\{S, Q\}$ of functions from R to the plane such that if t is in R then $K(\cdot, t)$ is in S and, for f in S , $Q(f, K(\cdot, t)) = f(t)$, it is necessary and sufficient that, for each function x from a finite subset M of R to the plane,

$$\sum_{\{s, t\} \in M \times M} x(s)[K(s, t)x(t)]^* \geq 0;$$

in case this latter condition is satisfied, there is only one such complete space $\{S, Q\}$: a function f from R to the plane belongs to S only in case there is a nonnegative β such that, for each function x from a finite subset M of R to the plane

$$\left| \sum_{t \in M} f(t)x(t)^* \right|^2 \leq \beta \left(\sum_{\{s, t\} \in M \times M} x(s)[K(s, t)x(t)]^* \right),$$

in which case $Q(f, f)$ is the least such number β .

Theorem 5.2 If $\{K_1, R, S_1, Q_1\}$ and $\{K_2, R, S_2, Q_2\}$ are kernel systems, then
(1) S_1 is a subset of S_2 only in case there is a nonnegative number β such that for each

function x from a finite set subset M of R to the plane

$$\sum_{\{s,t\} \in M \times M} x(s)[K_1(s,t)x(t)]^* \leq \beta \left(\sum_{\{s,t\} \in M \times M} x(s)[K_2(s,t)x(t)]^* \right),$$

and in this case the least such β is the least nonnegative number γ such that

$$Q_2(f, f) \leq \gamma Q_1(f, f) \text{ for each } f \text{ in } S_1.$$

(2) There is a kernel system $\{K_3, R, S_3, Q_3\}$ such that $K_3 = K_1 + K_2$, S_3 denotes the linear span $S_1 + S_2$ of S_1 and S_2 , and for $\sigma \in S_3$, $Q_3(\sigma, \sigma)$ is the minimum value of $Q_1(\sigma_1, \sigma_1) + Q_2(\sigma_2, \sigma_2)$ for all $\sigma_1 \in S_1$ and $\sigma_2 \in S_2$ with $\sigma = \sigma_1 + \sigma_2$.

(3) There is a kernel system $\{K_4, R, S_4, Q_4\}$ such that S_4 is the common part $S_1 \cap S_2$, $Q_4 = Q_1 + Q_2$, and (with Q_3 as in (2))

$$K_4(s, t) = Q_3(K_1(\cdot, s), K_2(\cdot, t)) = \frac{1}{4} \{ (K_1 + K_2)(s, t) - Q_3((K_1 - K_2)(\cdot, s), (K_1 - K_2)(\cdot, t)) \}$$

for $\{s, t\} \in R \times R$.

Terminology. If each of $\{K_1, R, S_1, Q_1\}$ and $\{K_2, R, S_2, Q_2\}$ is a kernel system, the parallel sum of K_1 and K_2 , $K_1 : K_2$, is the function K_4 described in Theorem 5.2(3), so that $\{K_4, R, S_4, Q_4\} = \{K_1 : K_2, R, S_1 \cap S_2, Q_1 + Q_2\}$.

Corollary to Theorem 5.2. If each of K_1, K_2, K_3 , and K_4 is an evaluation kernel in a complete inner product space of functions from R to the plane then so is the function $(K_1 + K_3) : (K_2 + K_4) - (K_1 : K_2 + K_3 : K_4)$.

Concluding Remarks

The reader may find several examples of such kernel-functions K from the earlier Chapters, 1-4, of this project. Professor Mac Nerney develops the idea of a kernel-space in considerably greater generality. The values of S are taken to be in a complete complex inner product space Y and the values of K to be in the space $L(Y)$ of bounded linear transformations on Y . As an historical note, the simplified version of Theorem 2 [1], offered here as Theorem 5.1 (Y the plane so that $L(Y)$ is also), has its origin in the first half of the Twentieth Century in the works of E. H. Moore, S. Bergman, and N. Aronszajn. Further historical detail may be developed by perusal of the References in [1].

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